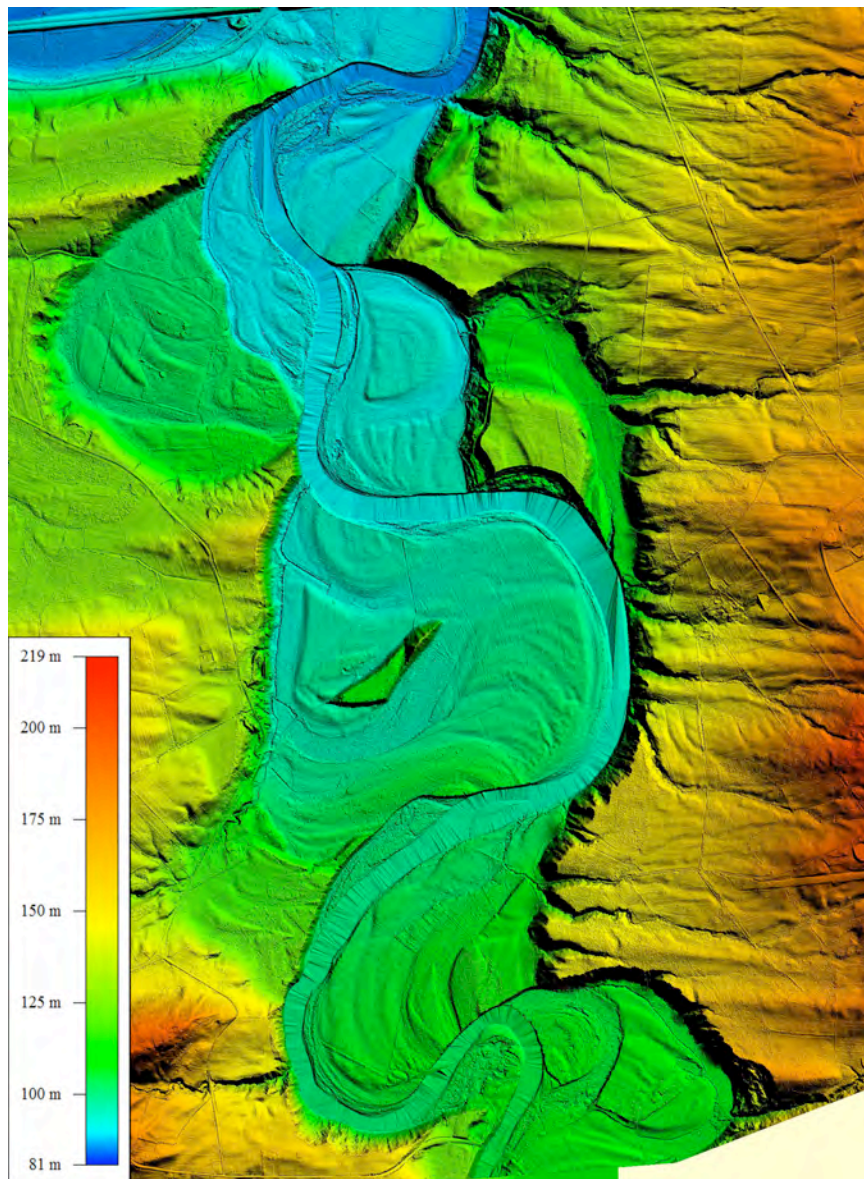


Mohawk Watershed Symposium

2010



Abstracts and Program

Olin Center, Union College
Schenectady NY
19 March 2010

DETERMINATION OF HISTORICAL CHANNEL CHANGES AND MEANDER CUT-OFF POINTS USING LiDAR AND GIS IN SCHOHARIE CREEK, NY

Marsellos, A.E.^{1,2}, Garver, J.I.², Cockburn, J.M.H.², Tsakiri, K.G.³

¹ *Dept. of Atmospheric & Environmental Sciences, State University of New York, Albany NY 12222, New York, U.S.A., email: marsellos@gmail.com*

² *Dept. of Geology, Union College, Schenectady, NY 12308, New York, U.S.A*

³ *Dept. of Mathematics & Statistics, State University of New York, Albany*

Introduction

The lower reach of the Schoharie Creek (Fig. 1) takes an unusual route through a glaciated bedrock high (now Lost Valley) and then a significant reach underlain largely by glacial till before the confluence with the Mohawk River. During the Recent evolution, this section of the river has incised downward, likely to keep pace with downward incision of the Mohawk River. The objective and the scope of this work is to utilize LiDAR data to construct a bare-earth model that allows identification of subtle terrain features such as abandoned channels and landslides. These geomorphological features reveal a complex history of incision and avulsion in the lower reaches of Schoharie Creek.

Bare-earth model

Light detection and Radar (LiDAR) is used to generate high-quality digital elevation data. These highly accurate topographic data can be used to analyze flood hazards, and to delineate floodplain boundaries because the topography is revealed in incredible detail. LiDAR sensors utilize a laser pulse (typically between 0.5 and 1 meter in diameter) and a pulse length (a short time of the laser pulse). LiDAR sensors are capable of receiving multiple returns, commonly up to five returns per pulse. Thousands of returns per second can be recorded classifying targets according to the number of returns. When a laser pulse hits a soft target (e.g., a forest canopy), the first return represents the top of that feature representing the top of this feature. However, a portion of the laser light beam likely continues downwards below the soft target and hit a tree branch or the ground below a tree. This would provide a second return. Theoretically, the last return represents the bare earth terrain. A classification of the points with the highest number of returns could reconstruct the ground surface (e.g. a TIN surface), while the rest of the points with lower number of returns could represent anthropogenic structures or

forest canopy. Surface water (lakes or rivers) does not return laser light and therefore a void is created that shows the outline of a current river channel or lakes.

Methods

The collected LiDAR data have a resolution of greater than 12 pts per m² providing a resolution in the gridded data at 0.25 m or less (commonly 0.09 m). The point cloud data has had minimal processing to eliminate outliers, from reflections etc. Larger areas with outliers or reflection have been manually subtracted and interpolated with the surrounding data points (this process mainly affects the river channel).

The high-resolution topographic images that show bare-earth LiDAR-derived topography are made by subtracting the canopy and defining a "bare-earth" elevation model using only the classified ground-points (Fig. 2) to identify evidence of the evolution of incision revealed by abandoned channels. Avulsion and subsequent abandonment of fluvial channels is analyzed by geomorphic mapping of these high-resolution topographic data (Fig. 3).

To facilitate viewing, interpretation and post-processing of the point cloud data 3D elevation models were constructed with examination of water levels in the TIN model (e.g. Fig 2). The flood plain area has been evaluated by the mapping highest elevated abandoned channels. A water level plain surface at 120 m in the TIN has specified as the TIN flooded area. The TIN flooded area has been converted to polygon features clipping the flood plain coverage. The path of the Schoharie Creek has been extracted by delineating the gap area from the LiDAR points. The abandon channel features have been extracted from the TIN model derived by the ground points. Those successive abandon channels have been classified according to the distance from the current creek location, and then connected to reconstruct historical pathways of the Schoharie Creek. All the extracted linear features from those historical

pathways have had same beginning and ending points, and their distance was measured.

Results

In Schoharie Creek, delineation of a series of successive abandon channels shows a clear evolutionary trend in this part of the river of successive channel formation and abandonment (Fig. 3, 4). There are at least three different channel deviation and incision times (Fig. 5). The succession of these channels helps to identify the principal trends (see Fig. 3) of the migration of meander bends that areas are currently used for agriculture. An important question that emerges from this analysis is the primary driving mechanism caused channel avulsion.

In the study area, the present current creek length is 9.0 km. The length measurements of the three successive channel deviations are 10.1, 11.2, and 12.2 km. In the same floodplain area the three reconstructed paleomeanders show a ~30% decrease of their length. The successive meander changes of their length (Fig. 5) show a linear association that may allow a broad prediction for the future meander length. This study is a first step in using high-resolution LiDAR data to Quantify Landscape evolution in the Mohawk watershed. Future analyses may include dating techniques to understand the temporal pace of these changes so that they may be linked to basin-wide processed.

Table 1: Meander length measurements (km) of different generations with the relative change (%). Prediction of the future has derived from a linear equation ($y = -1.06x + 13.31$).

Meander Generation	Length (km)	% change
C (older)	12.22	
B	11.2	8.35
A (younger)	10.17	9.20
Today	9.01	11.41
Future	7.99	11.38



Fig. 1: Aerial view from the study area of the Schoharie Creek (from Google Earth).

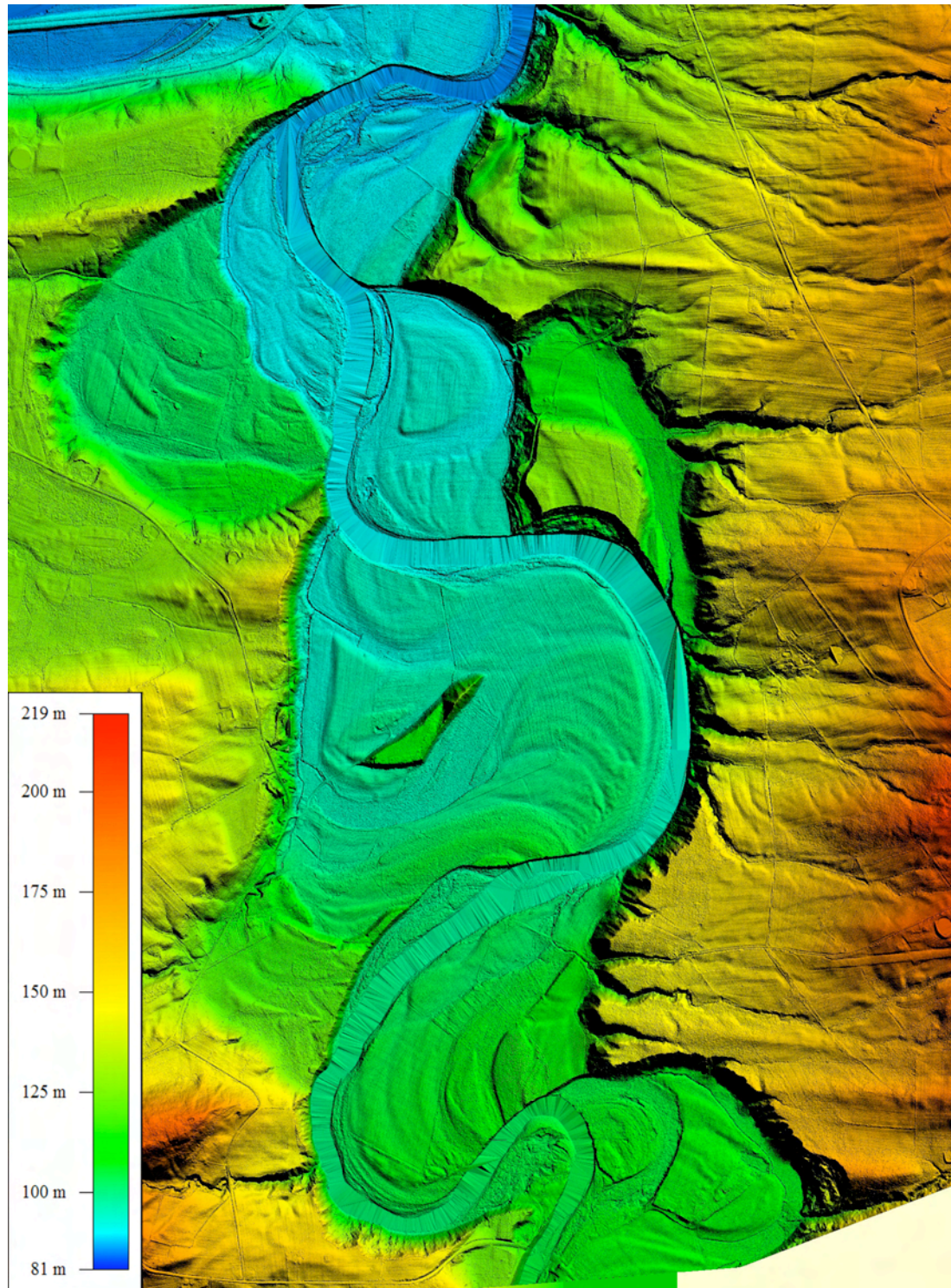


Fig. 2: TIN model derived from LiDAR ground points (bare-earth model)

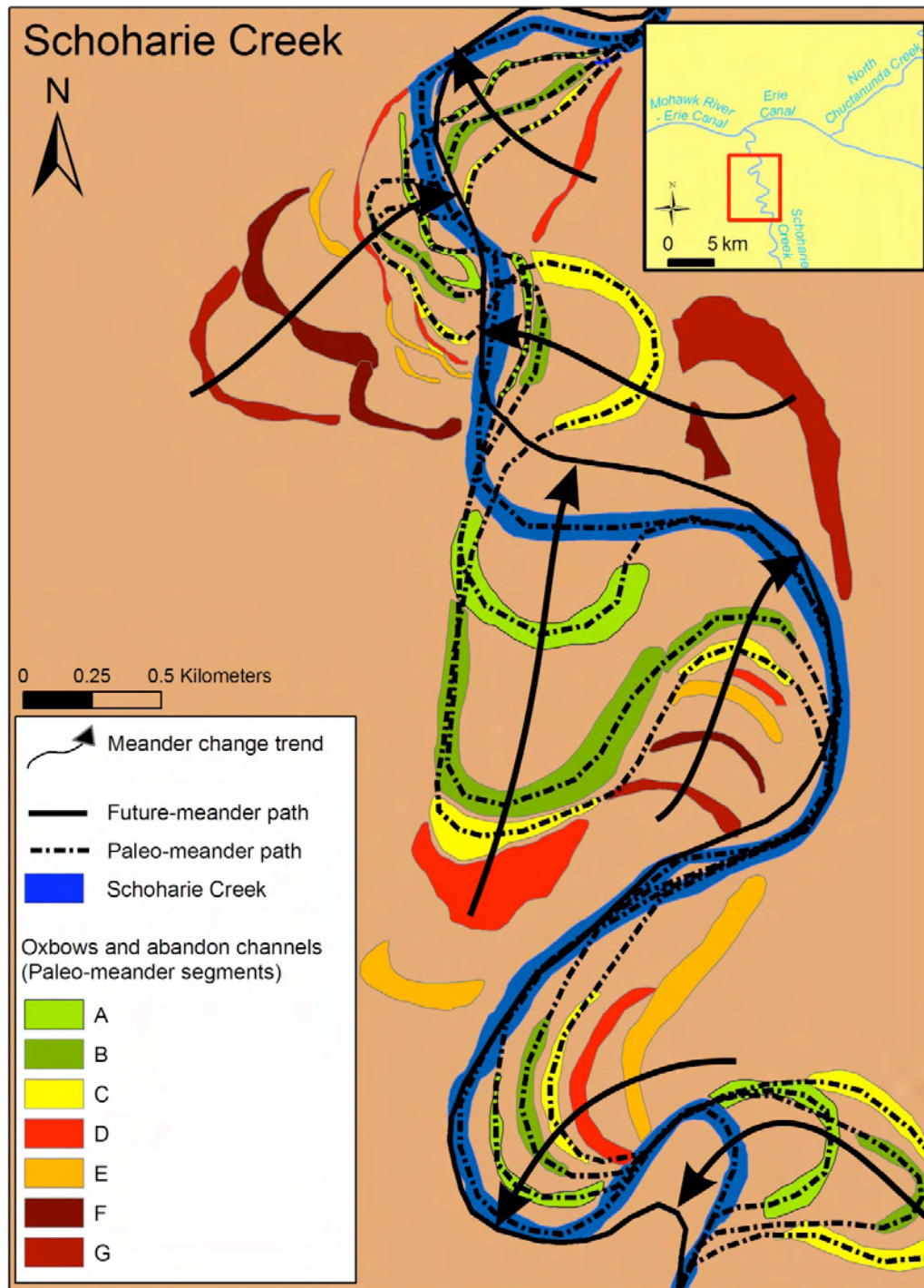


Fig. 3: Classification of oxbows and abandon channels as segments of older meanders.

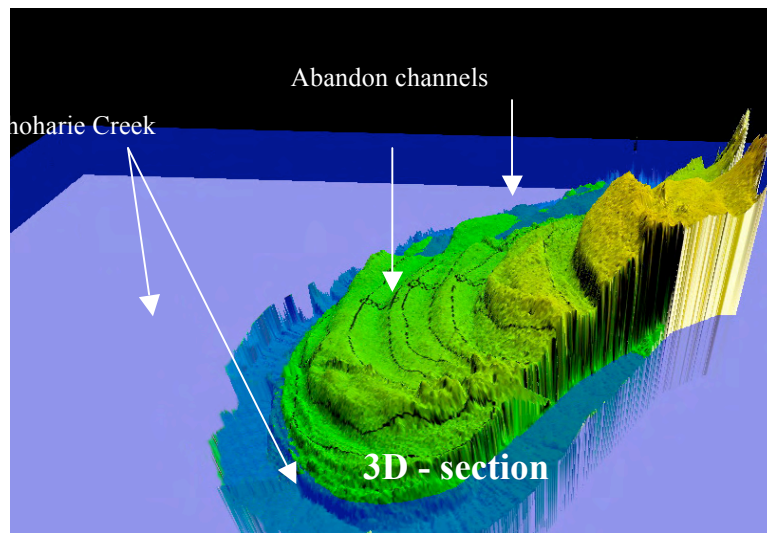


Fig. 4: A portion of the LiDAR bare-earth model that shows the successive abandon channels in 3D.

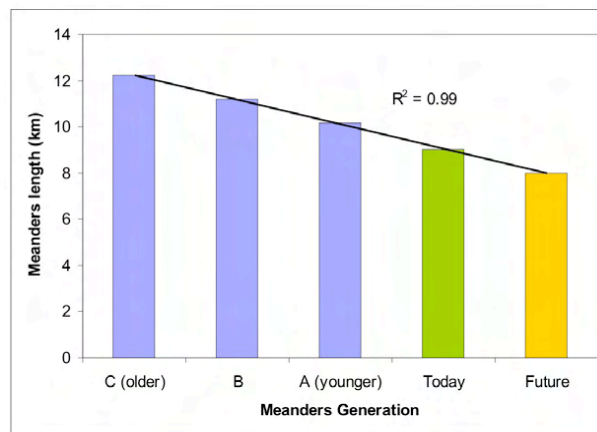


Fig. 5: The length of the present river path that appears to decrease by about ~10%.